

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0552

Airline 230 kV
December 31, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0552. The Interconnection Customer proposes a 1034.2 MW natural gas combined cycle facility interconnecting at the Airline 230 kV substation in the Ohio area. The request was received on December 16, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 194 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$49,556,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

DEVIL'S SLIDE - LUCENT 230.0 kV: 129.0% loading (600 MVA) following the outage of the OMNI-HACKBERRY - AIRLINE 230.0 kV line; DFAX = 66.2%.

The DEVIL'S SLIDE - DECORATIVE PANELS INTL 230.0 kV is overloaded to 127.4% of its 640 MVA emergency rating for the loss of the OMNI-HACKBERRY - AIRLINE 230.0 kV. The project contributes 66.2% DFAX (684.8 MW) to this overload.

The AIRLINE - DECORATIVE PANELS INTL 230.0 kV is overloaded to 125.4% of its 702 MVA emergency rating for the loss of the AIRLINE - OMNI-HACKBERRY 230.0 kV. The project contributes 66.2% DFAX (684.8 MW) to this overload.

For the loss of DECORATIVE PANELS INTL - AIRLINE 230.0 kV Line, the AIRLINE - OMNI-HACKBERRY 230.0 kV loads to 120.2% of its 850 MVA rating. Project DFAX: 82.6%.

For the loss of DECORATIVE PANELS INTL SUBSTATION - AIRLINE 230.0 kV, the OMNI-HACKBERRY - LUCENT 230.0 kV loads to 104.0% of its 653 MVA rating. Project DFAX: 19.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DEVIL'S SLIDE - LUCENT 230.0 kV	Rebuild line with higher-capacity conductor	\$9,143,000	100% (\$9,143,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DEVIL'S SLIDE - DECORATIVE PANELS INTL 230.0 kV line	Rebuild line with higher-capacity conductor	\$7,748,000	100% (\$7,748,000)
AIRLINE - DECORATIVE PANELS INTL 230.0 kV	Rebuild line with higher-capacity conductor	\$14,448,000	100% (\$14,448,000)
AIRLINE - OMNI-HACKBERRY 230.0 kV	Rebuild line with higher-capacity conductor	\$14,467,000	100% (\$14,467,000)
OMNI-HACKBERRY - LUCENT 230.0 kV	Reconductor line	\$4,118,000	91% (\$3,750,000)
	Total allocated to project		\$49,556,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.